



TWO-DAY NATIONAL CONFERENCE

Net-Zero Emission Technologies for Sustainable Development



Challenges and Opportunities

NOET-2027

JANUARY 29 – 30, 2027



CCUS · Green Hydrogen · Renewable Energy · Advanced Materials · AI for Energy Optimisation

CCUS & DAC

Green H₂

Renewables

Bioenergy

AI & Storage

Organized by

Department of Chemical Engineering
IIT (ISM) Dhanbad



ABOUT THE EVENT

About the Conference & Department



ABOUT THE CONFERENCE

Net-zero emissions mean balancing anthropogenic greenhouse-gas (GHG) emissions with their removal from the atmosphere — a balance essential for sustainable development and for limiting global warming. The transition to a net-zero future demands transformative technologies that decarbonize energy, industrial and transportation systems while minimizing CO₂ emissions. Emerging solutions such as **Carbon Capture, Utilization & Storage (CCUS), Direct Air Capture (DAC), green hydrogen, renewable energy, bioenergy, advanced energy storage, separation processes, and AI for energy optimization** are expected to play a crucial role.

Significant technical, economic and policy challenges remain — calling for interdisciplinary research and strong collaboration among academia, industry and government. NOET-2027 provides a platform for researchers, academicians, industry professionals, policymakers and students to exchange ideas and explore innovative solutions for a low-carbon future, supporting India's **Net-zero 2070** commitment and the Paris Agreement goal of **1.5°C**.

2070

India's Net-Zero Target Year

1.5°C

Paris Agreement Goal

2 Days

Keynote, Invited & Poster Sessions

ABOUT THE DEPARTMENT

Since its inception in **2010**, the Department of Chemical Engineering has aimed to instill in every student the essential skills of critical thinking and creative problem-solving needed for success in the practice of chemical engineering. The department offers academic and research programs leading to **B.Tech, M.Tech and PhD** degrees. Many graduates are placed in reputed national and multinational organizations, and a substantial number pursue higher studies and research at premier institutes in India and abroad. The department is actively engaged in collaborative, multidisciplinary research sponsored by CIL, DST, DST-FIST, ANRF, SAIL, TATA Steel, NJB, TEXMiN and others — continually modernizing its research infrastructure and strengthening cooperative links with industry, R&D houses, government departments and premier academic institutions for the advancement of knowledge and the benefit of society.

GRADUATES RECRUITED BY

IOCL

BPCL

ONGC

GAIL

Reliance Industries

L&T

Vedanta

Haldia Petrochemicals

Aditya Birla

8 MAJOR THEMES

CCUS · Green Hydrogen · Renewables · Energy Storage · Materials · Catalysis · Circular Economy

Full details, descriptions & conference programs overleaf → see page 3



SCOPE

Major Themes & Conference Programs



MAJOR THEMES



01 · Clean Energy Innovation

Next-gen technologies driving the low-carbon transition



02 · CCUS

Carbon Capture, Utilization & Storage



03 · Green Hydrogen

Production, storage, transport & utilization



04 · Renewable Energy

Solar, wind, hydro & geothermal systems



05 · Energy Storage Systems

Batteries, supercapacitors & flow batteries



06 · Materials & Nanotechnology

Nanomaterials, catalysts & functional materials



07 · Catalysis

Electro-, Chemical, Bio- & Photo-catalysis pathways



08 · Circular Economy

Resource efficiency & waste valorization



09 · Other Relevant Areas

Interdisciplinary contributions welcome

CONFERENCE PROGRAMS



Keynote Talks

Distinguished experts sharing visionary insights on the future of net-zero technologies.



Invited Talks

Focused sessions by leading researchers from academia and industry.



Oral / Poster Presentations

Original research contributed and presented by participants.



PARTICIPATE

Call for Abstract, Key Dates & Fees



CALL FOR ABSTRACT

The conference invites abstracts from faculty members, students, scientists, researchers and industry professionals. Abstracts should be **200–250 words**, reporting original research on any of the major themes or related areas. All submitted abstracts will be subject to **peer review**.

Correspondence & submissions: n0et@iitism.ac.in | Guidelines: [NOET-2027 abstract portal link](#)

IMPORTANT DATES

5 Sep 2026	Commencement of Abstract Submission
27 Nov 2026	Abstract Submission Deadline
7 Dec 2026	Acceptance of Abstract
8 Dec 2026	Commencement of Registration
4 Jan 2027	Full-length Manuscript Deadline
20 Jan 2027	Online Registration Deadline
29–30 Jan 2027	Conference Dates

REGISTRATION FEES

Type of Participant	Regular Fee
Industrial Personnel	INR 12,000
Faculty Member & Scientist	INR 8,000
Student & Research Scholars	INR 3,000
Others	INR 8,000

Sponsorship: Opportunities available. See NOET-27 Sponsorship link for rates & benefits.

WHY ATTEND



NETWORK

Connect with academia, industry & policymakers



PUBLISH

Present peer-reviewed research on a national platform



LEARN

Gain insights from keynote & invited experts



COLLABORATE

Explore joint research & industry linkages



PEOPLE

Patrons & Organizing Committee

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Prof. Sukumar Misra

Director, IIT (ISM) Dhanbad

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SD

Prof. Sarat Kumar DasDy. Director, IIT (ISM)
Dhanbad**CHAIRMAN**

AK

Prof. Aditya Kumar

Head, Chemical Engineering

LOCAL ORGANIZING COMMITTEE Convener: **Prof. Paidinaidu Paluri** | Co-Conveners: **Prof. Arunkumar Samanta & Prof. Lutukurthi D N V V Konda**

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CONTACT US**CONVENER****Prof. Paidinaidu Paluri**naidu@iitism.ac.in
+91-9471192195**CO-CONVENER****Prof. Arunkumar Samanta**asamanta@iitism.ac.in
+91-9471192213**CO-CONVENER****Prof. Lutukurthi D N V V Konda**dnvkvkonda@iitism.ac.in
+91-9471192194



JOIN US

Registration Form & Payment Details



Two-Day National Conference on Net-Zero Emission Technologies for Sustainable Development: Challenges and Opportunities (N0ET-2027)

January 29–30, 2027

Name	
Qualification	
Designation	
Department	
Institution / Organisation	
Official Address (with Pin)	
Phone	
Email	
Accommodation Required (Yes/No)	

PAYMENT DETAILS

Mode of Payment (DD / Payee Cheque / Online)	
Amount Paid	
Date of Payment	
Cheque No. / DD / Transaction ID	

BANK ACCOUNT DETAILS · SCAN & PAY

ACCOUNT INFORMATION

Name of Account **IIT ISM CEP ACCOUNT**
Account Number **110261358281**
Name of the Bank **CANARA BANK**
IFS Code **CNRB0000986**
SWIFT Code **CNRBINBBFD**



Please mention **"OCEP 0137 O & N0ET-2027"** while making the transaction.

Date

Signature of the Applicant

How to submit: Complete this form, attach proof of payment, and email to n0et@iitism.ac.in, or submit online via the N0ET-27 registration portal.